

COURSE CODE/TITLE:	CPE 010	SECTION:	CPE21S2
DATE PERFORMED:	July 8, 2026	DATE SUBMITTED:	July 10, 2026
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ACTIVITY NO. 1.1 Algorithm, Flowchart, Pseudocode

1. PROCEDURE OUTPUT

TASK 1

Algorithm

1. Declare the user's option.
2. Declare the scores of the quizzes, total score, and average using "double."
3. Code structure displays the menu.
4. Asks the user to enter the quiz scores of the students.
5. The user enters the quiz scores.
6. Code structure computes the total score and average.
7. Code structure displays the total score and average of the students.
8. Code structure ends.

Pseudocode

BEGIN

DECLARE Student1, Student2, Student3, total, average

PRINT "=====

PRINT " Task 1 Grade Computation "

PRINT "=====

PRINT "Enter the quiz scores of the students."

INPUT Student1, Student2, Student3,

total = Student1 + Student2 + Student3

average = total / 3.0

PRINT "=====

PRINT " Results: "

PRINT "=====

PRINT "Total score of quizzes" , total

PRINT "Average score of quizzes", average

END

```

G+ main.cpp •
Seatwork 1 > G+ main.cpp > main()
7   int main() {
8       cout << "===== " << endl;
9       cout << "  Task 1 Grade Computation" << endl;
10      cout << "===== " << endl;
11      cout << endl;
12      cout << "Enter the quiz scores of the students.";
13      cin >> Student1 >> Student2 >> Student3;
14      total = Student1 + Student2 + Student3;
15      average = total / 3.0;
16      cout << "===== " << endl;
17      cout << "  Results:" << endl;
18      cout << "===== " << endl;
19      cout << "Total Score of quizzess: " << total << endl;
20      cout << "Average Score of quizzess: " << average << endl;
21
22      return 0;
23  }

```

Code snippet 1: Task 1 Grade Computation

Analysis of code snippet 1:

- I designed my code structure first before proceeding to the processing of computation. I also made my code structure where it has a user input and used basic arithmetic operations to compute the total score of the students and also its average. After computing the scores and average of the quiz, the code structure successfully displayed the total score and average of the quizzes.

```

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
● lourdes@Dhesss-MacBook-Air Seatwork 1 % "/Users/lourdes/Desktop/CPE010_SILA0/Seatwork 1/main"
=====
  Task 1 Grade Computation
=====

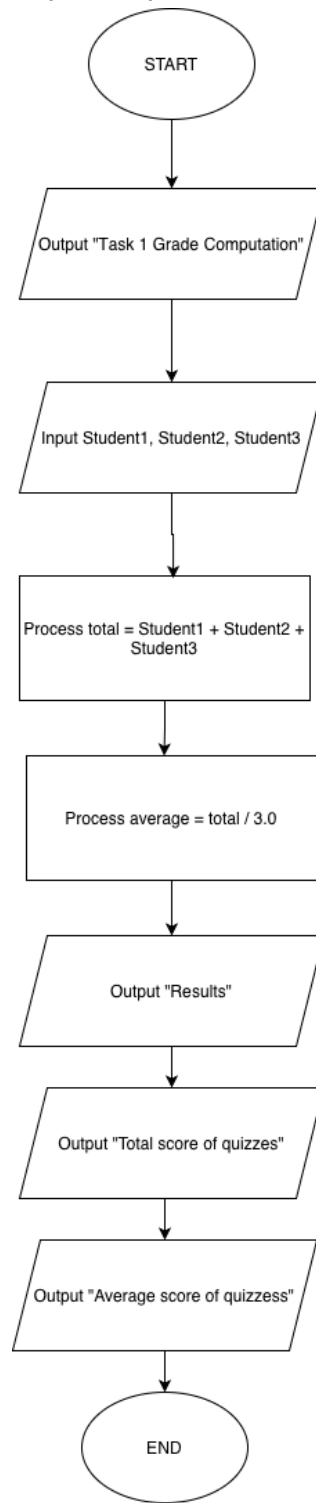
Enter the quiz scores of the students.12.4 1 100
=====
  Results:
=====
Total Score of quizzess: 113.4
Average Score of quizzess: 37.8
○ lourdes@Dhesss-MacBook-Air Seatwork 1 %

```

Output snippet 1: Task 1 Grade Computation

Analysis of output snippet 1:

- This code structure successfully displayed the expected output of the program.

**Flowchart snippet 1: Task 1 Grade Computation**

Analysis of flowchart snippet 1:

- In this flowchart snippet, I used the basic properties of the flowchart such as the rectangle for processing, parallelogram for input/output, and lastly the oval for the start/end. I also made it simple and short to easily understand it.

TASK 2

Algorithm

1. The user gathers the items.
2. The user comes to the cashier.
3. User inputs the price of Item 1, Item 2, and Item 3.
4. Code structure computes the total price of 3 items.
5. Code structure displays the total price.
6. The user inputs the amount to pay.
7. Code structure computes the change.
8. Code structure displays the official receipt.

Pseudocode

BEGIN

DECLARE Item1, Item2, Item 3, total, payment, change

PRINT "Task 2 (Cashiering System)"

PRINT "Enter the price of Item 1:"

INPUT Item1

PRINT "Enter the price of Item 2:"

INPUT Item2

PRINT "Enter the price of Item 3:"

INPUT Item3

total = Item1 + Item2 + Item3

PRINT "Total Price:"

PRINT "Enter amount to pay: "

INPUT payment

change = payment - total

PRINT "=====

PRINT " OFFICIAL RECEIPT"

PRINT "=====

PRINT "Price of Item 1: " " Php "

PRINT "Price of Item 2: " " Php "

PRINT "Price of Item 3: " " Php "

PRINT "=====

PRINT "Total Price: " " Php "

PRINT "Payment: " " Php "

PRINT "Change: " " Php "

PRINT "=====

```
main.cpp X
main.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  double Item1, Item2, Item3;
5  double total, payment, change;
6  int main(){
7      cout << "Task 2 (Cashiering System)" << endl;
8      cout << "Enter the price of Item 1:" << endl;
9      cin >> Item1;
10     cout << "Enter the price of Item 2:" << endl;
11     cin >> Item2;
12     cout << "Enter the price of Item 3:" << endl;
13     cin >> Item3;
14
15     total = Item1 + Item2 + Item3;
16     cout << "Total Price: " << total << endl;
17
18     cout << "Enter amount to pay: ";
19     cin >> payment;
20
21     change = payment - total;
22
23     cout << "=====" << endl;
24     cout << "  OFFICIAL RECEIPT" << endl;
25     cout << "=====" << endl;
26     cout << "Price of Item 1: " << Item1 << " Php " << endl;
27     cout << "Price of Item 2: " << Item2 << " Php " << endl;
28     cout << "Price of Item 3: " << Item3 << " Php " << endl;
29     cout << "=====" << endl;
30     cout << "Total Price: " << total << " Php " << endl;
31     cout << "Payment: " << payment << " Php " << endl;
32     cout << "Change: " << change << " Php " << endl;
33     cout << "=====" << endl;
34
35     return 0;
```

Code Snippet 2: Task 2 Cashiering System

Analysis of code snippet 2:

- I first declared the items, payment, total price and the change. Made a user input for the price of the items. Used basic arithmetic operations to compute the total price of the user's items and change of the user. After computing the total price and change of the user, the code structure will display the official receipt of the user and also display the

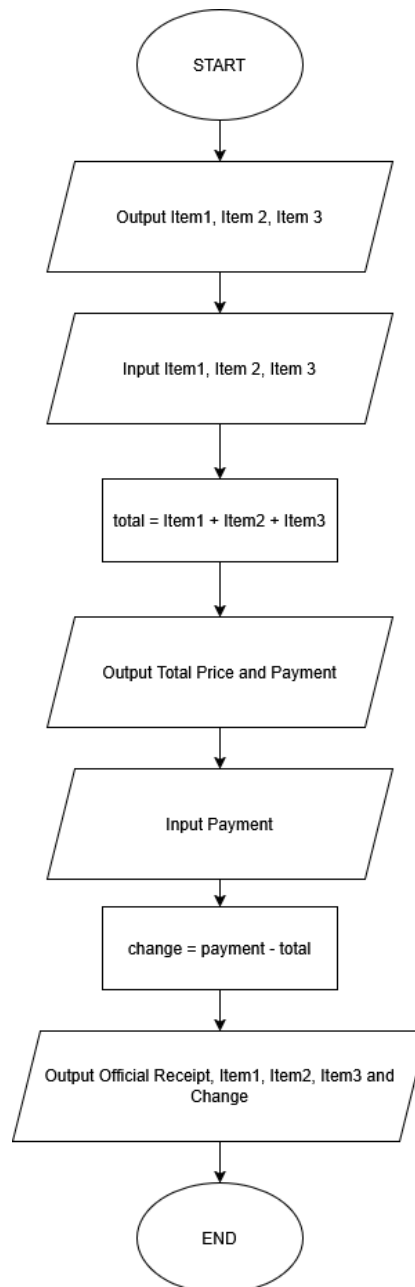
price of the items that the user inputted. I also made some designs to make the program look organized.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
=====
PS C:\Users\TIPQC\Documents\CPE010_Silao\Seatwork 1> ^C
PS C:\Users\TIPQC\Documents\CPE010_Silao\Seatwork 1>
PS C:\Users\TIPQC\Documents\CPE010_Silao\Seatwork 1> & 'c:\Users\TIPQC\.vscode\extensions\ms-vscode.cpptools-1.32.2-win32-x64\debugAdapters\bin\Windows
Debuglauncher.exe' '--stdin=Microsoft-MIEngine-In-eikjheqv.ecq' '--stdout=Microsoft-MIEngine-Out-uve30rcm.rnz' '--stderr=Microsoft-MIEngine-Error-geejcj
nu.rvq' '--pid=Microsoft-MIEngine-Pid-udrt21pj.b3w' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Task 2 (Cashiering System)
Enter the price of Item 1:
12.5
Enter the price of Item 2:
125.6
Enter the price of Item 3:
123.5
Total Price: 261.6
Enter amount to pay: 261.6
=====
OFFICIAL RECEIPT
=====
Price of Item 1: 12.5 Php
Price of Item 2: 125.6 Php
Price of Item 3: 123.5 Php
=====
Total Price: 261.6 Php
Payment: 261.6 Php
Change: 0 Php
=====
PS C:\Users\TIPQC\Documents\CPE010_Silao\Seatwork 1> |
```

Output Snippet 2: Task 2 Cashiering System

Analysis of output snippet 2:

- In this output snippet, it shows that the program successfully display the expected output.



Flowchart Snippet 2: Cashiering System

Analysis of flowchart snippet 2:

- In this flowchart snippet, I used the basic properties of the flowchart such as the rectangle for processing, parallelogram for input/output, and lastly the oval for the start/end. I also made it simple and short to easily understand it.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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